Exercise Session – Chapter 3: Quality of Service

Exercise 1:

Explain why QoS is necessary in modern networks. Give three real-world examples where QoS is critical and describe what happens if QoS is not implemented.

Exercise 2:

Define the following parameters and explain how each one affects user experience:

- Bandwidth
- Delay
- Jitter
- Packet loss
  Provide one example of an application sensitive to each.

Exercise 3:

Briefly describe the token bucket algorithm and its role in traffic shaping.
Then, suppose a bucket has a capacity of 10 tokens and a rate of 2 tokens/second. How long will it take to refill after a burst of 10 packets?

Exercise 4:

Given the following DSCP values, identify the type of traffic and priority:

1. 46
2. 0
3. 10
4. 34

Exercise 5:

Why do QoS mechanisms often depend on **Layer 3 and 4 information** (IP address, port number) rather than **Layer 2 (MAC address)**?

CamScanner